

Testing Concept Brief

Matthew Santorsa

Definitions

- **Exhaustive Test Generation:** The process of applying every possible input combination (test vector) to a circuit to guarantee that all detectable faults are observed. For an n -input circuit, this requires 2^n test vectors.
- **Fault-Detection Matrices:** A 2D grid mapping test vectors to the specific faults they detect. Rows represent test vectors, columns represent potential faults, and a cell contains a 1 if the vector detects the fault while a 0 if not.
- **Fault Coverage:** Representation of the proportion of total possible faults that are successfully detected by a given set of test vectors. It is calculated as $(\text{Detected Faults} / \text{Total Faults}) \times 100$
- **Set-Cover Interpretation:** Framing the testing process as a mathematical set-cover problem, where the universe is the total set of faults, and each test vector represents a subset of detectable faults. The goal is to find the minimum number of subsets that completely cover the universe.
- **Greedy Selection:** Heuristic optimization algorithm that builds a solution step-by-step by always choosing the next test vector that will detect the largest number of previously uncovered faults, without reconsidering past choices.

Worked Example

Below is a table depicting a 2 input AND gate, which shows faulty versus correct outputs to help identify successful test vectors. The

setup is 2 inputs, A and B, with 1 output Z. The injected fault is a stuck at 1 fault, which implies the gate will always behave as if input A is receiving a 1.

Test Vector (a,b)	Correct Output (z)	Faulty Output (Z_faulty)	Fault Detected?
V1: (0,0)	0	0	No
V2: (0,1)	0	1	Yes (inconsistent outputs)
V3: (1,0)	0	0	No
V4: (1,1)	1	1	No

We can conclude that the only successful vector is vector 2, as it is the only vector capable of showing the fault altering logic output.